

AI531C — Week 7 Pre-Class Activity (PCM-7: Shape Surgery)

Prepares: Lecture 17 (Morphology & descriptors, Tue 15 Sep) · Lectures 18–19 (Classical fusion & change detection; Limits of classical pipelines), Wed 16 Sep · Tutorial T7 (Tue 15 Sep).

1 · Topic

Erosion, dilation, opening, closing: shape surgery with a structuring element — and the measurements (area, centroid, bounding box) that follow.

2 · Why you are preparing this

Lecture 17 treats morphology as a precision instrument, not a magic cleanup. Arriving with the erosion arithmetic done lets class explore the operations' ALGEBRA — why opening $\circ$ opening = opening, and when closing invents bridges that were never there.

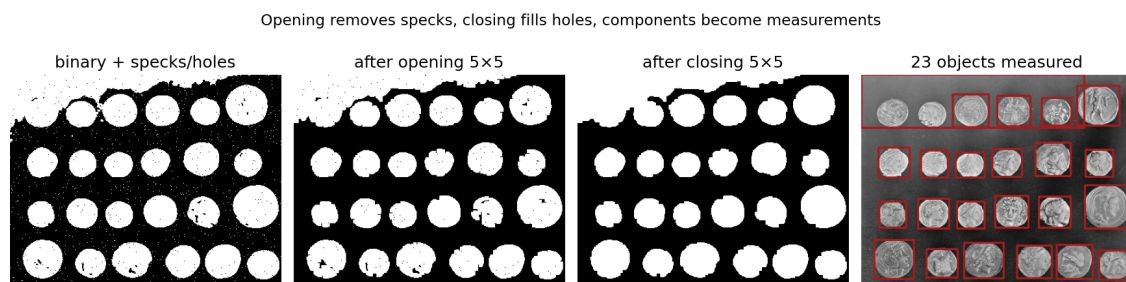
3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 17 chapter, the sections on the four operations and on connected-component statistics (about 3 pages);
- Lecture 19 chapter, ONLY its first page ('where classical pipelines crack' — Unit III's doorway).

Optional depth: GW4 Ch. 9, Secs. 9.1–9.3 (the SE figures).

4 · Interpret one figure (4 min)



Source: course figure, Lecture 17 (morphology operations panel).

In two sentences: which operation removed the speckles and which closed the holes — and why does the ORDER (open then close, vs close then open) matter?

5 · One small calculation (5 min)

A solid 10×10 foreground square is eroded by a 3×3 square SE. Give the output size and the area lost. What happens if you erode a 2-pixel-wide line with the same SE?

6 · Predict (4 min)

Predict: after opening with a 5×5 SE, can any foreground object SMALLER than the SE survive? Commit to yes/no with one line of reasoning.

7 · Inspect this fragment (3 min)

```
clean = cv2.morphologyEx(bw, cv2.MORPH_OPEN, k)
n, lab, stats, cents = cv2.connectedComponentsWithStats(clean)
print(stats[1:, cv2.CC_STAT_AREA]) # why drop row 0?
```

Question: what does row 0 of stats describe, and why is it excluded before measuring objects?

8 · Bring one question to class

One question about morphology — or one binary-image cleanup problem from your own domain where you suspect the SE size is the whole game.

Submit: items 4-6 — half a page maximum, before Tue 15 Sep, 9:00 AM.

Expected time: 30-35 minutes. Counts toward the weekly preparation portfolio; only this card is submitted (per-lecture primers are optional). Note: Assessment Test I runs this period per the lesson plan — this task is deliberately lighter.